BIO203

GDB

Spring 2021

We Grow Together

PCR and Southern Blotting are both types of recombinant DNA technology.

PCR stands for "polymerase chain reaction" in which a DNA sequence is amplified up to a million fold. PCR utilizes heat treatment to unwind DNA, along with primers and polymerization. It is a fast process and no vectors are required making it useful in the sequencing of DNA.

This is especially helpful forensically when only limited DNA may be available. It is also useful in the detection of genetic disorders.

Southern Blotting is a method where DNA is cut with restriction enzymes and then probed with DNA. In a simplified explanation, first the DNA is separated by gel electrophoresis, then it is transferred to a solid support such as nitrocellulose paper. After that the probe is added and studied by means of radiography or enzyme reaction.

Summary

PCR and Southern Blotting are alike because they are both methods used in recombinant DNA

technology.
PCR and Southern Blotting are different because:
a) They use different methodology
b) PCR uses one reaction vessel/ Southern Blotting requires multiple vessels.
c) PCR is easy and fast/ Southern Blotting is more difficult and requires more time.
d) PCR is able to use very minute DNA samples/ Southern Blotting needs larger sequences of DNA